

PENCHALA ESWAR POTLURU

Email: eswarp2004@gmail.com | Phone Number: 6303011329 | Bangalore

LinkedIn: www.linkedin.com/in/penchala-eswar-potluru | Git Hub : <https://github.com/Eswarp2004>

CAREER OBJECTIVE

To build a career as an Embedded Engineer by leveraging my skills in C, C++, data structures, and system-level programming to design efficient and reliable embedded systems. I aim to contribute to innovative hardware-software integration and real-time applications while continuously advancing my knowledge in embedded technologies.

WORK EXPERIENCE

- Currently undergoing a technical training program – **Advanced Embedded Systems Course** at Emertxe Information Technologies (<http://www.emertxe.com>) Bangalore. This course is Government of India certified program, aligned with Skill India / NSDC under Electronics Sector Skill Council of India (<http://www.essc-india.org>) - **Embedded Software Engineer** QP ELE /Q1501

TECHNICAL SKILLS

- Programming Languages:
 - Advanced C programming
 - OOP using C++
 - Data structures
- Embedded Platforms:
 - Distributions – Linux (Ubuntu)
 - PIC Microcontroller (PIC16F877A)
- Embedded controllers:
 - Hands-on working with GPIOs, Analog I/Os, Memory usage, interfacing, character LCD
 - Peripherals usage - Timers, Counters and Interrupts
 - Communication protocols - UART, SPI, I2C, CAN etc
- Development Environment and Tools:
 - Development environment – MPLAB
 - Compilers – GCC, XC8, ARM-Linux-gcc
- System Programming
 - IPC mechanisms – Pipe, FIFO, Shared memory
 - Network programming using TCP and UDP sockets
 - Linux kernel system call

EDUCATION

- B.E. (ECE) – Sathyabama University – 7.36 CGPA – 2021 – 2025
- Class XII – 856 Marks – 2021
- Class X – 7.80 CGPA – 2019

PERSONAL ATTRIBUTES

- Adaptable to new technologies and quick at grasping complex technical concepts.
- Effective communicator who values teamwork, clarity, and knowledge sharing in collaborative environments.
- Committed to continuous learning and delivering reliable, high-quality work.

PROJECTS AT EMERTXE

Project Number:1

Title	Car Black Box
Git Hub URL	https://github.com/Eswarp2004/car_black_box_project.git
Project brief	This project involves the development of a Car Black Box system designed to monitor and log critical vehicle data in real time using the I ² C communication protocol. Various vehicle parameters are captured through sensors and synchronized with precise timestamp information provided by a Real-Time Clock (RTC). The collected data is then stored in an external EEPROM, ensuring non-volatile and secure data retention. During accidents or abnormal vehicle behavior, the system preserves vital records, enabling effective post-event analysis and investigation.
Technologies used	PIC16F877A Microcontroller, I ² C communication Protocol, UART protocol, RTC, EEPROM, Embedded C
Key challenges & Learnings	✓ Handling non-volatile EEPROM memory constraints by organizing and storing multiple event records effectively. ✓ Developing an event logging mechanism by integrating sensors, keypad, and display modules using Embedded C in a real-time system. ✓ Implementing I²C-based communication to interface multiple peripherals such as RTC and EEPROM over a shared bus efficiently.

Project Number:2

Title	Lexical Analyzer
Git Hub URL	https://github.com/Eswarp2004/lexical_analyzer.git
Project brief	The Lexical Analyzer is a program that converts a stream of characters into lexical tokens such as keywords, identifiers, and symbols. This project processes a C source file to generate tokens for the next stage of compilation, while handling possible errors during tokenization.
Technologies used	C Programming, Compiler Design Concepts, Lexical Analysis, Tokenization Techniques, File Handling, String Processing Functions (FSA), Standard C
Key challenges & Learnings	✓ Designing a lexical analysis mechanism to efficiently scan and classify source code into tokens such as keywords, identifiers, constants, operators, and symbols. ✓ Implementing file handling techniques in C to read and process source code files line by line for token generation. ✓ Developing logic to differentiate between reserved keywords and user-defined identifiers using string comparison and pattern matching techniques. ✓ Managing symbol table entries and token storage effectively to support further stages of the compiler design process.

Project Number:3

Title	MP3 Tag Reader
Git Hub URL	https://github.com/Eswarp2004/mp3-tag-reader.git
Project brief	MP3 tag reader is a software which will read and display MP3 (ID3) tag information from MP3 files. The software will be desktop based and not web based. This solution will read the given MP3 file, extract various tag information and display them via command line. This project can be extended to implement a tag editor, wherein users can modify MP3 tag information.
Technologies used	Advanced C – File handling, Pointers, String operations, Bitwise operations, Functions, Command line arguments
Key challenges & Learnings	✓ Faced difficulty parsing variable-length ID3v2 tag frames; resolved using frame header analysis and accurate offset calculations ✓ Encountered program crashes when reading multi-byte fields, solved by ensuring correct byte order conversion (big-endian to little-endian) and proper data type handling. ✓ Strengthened skills in binary file parsing, hexadecimal data interpretation, and methodical debugging approaches in C programming

Project Number:4

Title	Inverted Search Engine
Git Hub URL	https://github.com/Eswarp2004/Inverted-search-engine.git
Project brief	the objective was to build an inverted index to map words to their occurrences across multiple documents. Parsed each file to extract words, then stored their locations using a custom indexing structure for efficient retrieval. The final output allows fast and accurate full-text searches across the entire document set.
Technologies used	Advanced C – Structures, Linked Lists, Hashing, Pointers, Dynamic Memory Allocation, File Handling, Functions
Key challenges & Learnings	✓ Struggled with memory leaks and segmentation faults; resolved by carefully tracking dynamic memory usage and performing step-by-step debugging. ✓ Encountered repeated entries during word parsing across multiple files; resolved by integrating a duplicate-checking mechanism within the data insertion logic. ✓ Faced challenges in maintaining accurate word-to-file mappings across multiple files; solved by building a structured node system to track file occurrences and frequencies.

Project Number:5

Title	Arbitrary Precision Calculator
Git Hub URL	https://github.com/Eswarp2004/Arbitrary-Precision-Calculator-APC-.git
Project brief	The objective of this project is to perform basic mathematical operations such as addition, subtraction, multiplication, and division for very large numbers. The project focuses on developing efficient algorithms to handle these large numbers accurately, ensuring that the results are precise regardless of their size. This capability is important in areas like cryptography, scientific computing, and any application requiring high-precision calculations.
Technologies used	Advanced C – Linked Lists, Structures, Pointers, Dynamic Memory Allocation, File Organization, Functions, Command Line Arguments (argc, argv)
Key challenges & Learnings	✓ Implemented string manipulation (pattern matching) using library functions; faced challenges in selecting correct parameters. ✓ Faced challenges with memory deallocation using free() during linked list implementation. ✓ Faced challenges in function pointer declaration and correct address assignment, leading to segmentation faults.

Project Number:6

Title	Image Steganography using LSB Encoding and Decoding
Git Hub URL	https://github.com/Eswarp2004/lbs-steganography.git
Project brief	The objective was to hide secret information within a BMP image file without altering its visible properties. This was achieved by embedding the message into the least significant bits (LSB) of the image data, along with encoding the length of the message for accurate retrieval. The sender and receiver use a shared secret or magic string to extract the original data, enabling secure communication through seemingly ordinary images.
Technologies used	Advanced C – file operations, pointers, bitwise operations, dynamic memory, Makefiles, command-line arguments
Key challenges & Learnings	✓ Ensuring the embedded data did not alter the visible properties of the image; solved by carefully manipulating only the least significant bits ✓ Allowed flexible user input for source image, secret message file, and output image, improving usability and automation capability. ✓ Faced difficulties with shifting and masking bits correctly, which was resolved through practice and debugging. ✓ Handled encoding and decoding of message length and magic string to ensure accurate and secure data extraction

Project Number:7

Title	
Address Book	
Git Hub URL	https://github.com/Eswarp2004/AddressBook_project.git
Project brief	The Address Book is a small application written in C language. It keeps track of names and telephone/mobile numbers and e-mail addresses. It is a console-based application which uses standard I/O for adding and deleting contact names, phone numbers and e-mail addresses, searching names and associated numbers and email addresses, updating numbers and email addresses, and deleting contacts.
Technologies used	Advanced C – File operations, Pointers, Functions, Dynamic memory management
Key challenges & Learnings	✓ Encountered issues with duplicate contact names while performing search, delete, and update operations; overcame this by implementing additional checks using phone numbers and email addresses for unique identification ✓ Initially faced segmentation faults due to improper pointer handling; overcame this by initializing pointers properly and validating memory allocations before use. ✓ Developed a strong practical understanding of file handling, pointers, and command-line arguments while implementing the program in C.

ACADEMIC PROJECTS

Title	
Autonomous Navigation and Scanning System	
Project brief	This project provides a practical implementation of autonomous robotics using embedded systems, computer vision, and ROS-based simulations. It offers hands-on exposure to real-world robotic navigation, sensor integration, and intelligent path planning, making it highly suitable for academic research, embedded systems learning, and robotics development.
Technologies used	ESP32-CAM,GPS Module, DC Motors, Wi-Fi Communication
Key challenges & Learnings	✓ Developed an autonomous robot capable of obstacle detection and automatic path correction using ESP32-CAM and Robot Operating System. ✓ Controlled DC motors using the L298N Motor Driver Module to achieve smooth movement and direction control. ✓ Integrated GPS-based tracking and autonomous navigation for location monitoring and route adjustment. ✓ Gained experience in robotic control systems, sensor integration, path planning, and autonomous navigation techniques.